

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name UTKARSH GUPTA

Enrollment No. 2401020065

Jaypee Institute of Information Technology, Noida

End Term Examination, Odd 2024

B.Tech Semester: 1st

Course Name: Software Development Fundamentals-I

Maximum Marks: 35

Course Code: 15B11CI111

Maximum Time: 2 Hrs

Course Outcomes		Cognitive Levels
CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts	Understand (Level 2)
CO2	Explain basics of C programming concepts to make decision for solving problems	Understand (Level 2)
CO3	Demonstrate and contrast different methods for writing modular programs in C	Understand (Level 2)
CO4	Use various C programming constructs to implement iteration and recursion	Apply (Level 3)
CO5	Apply and implement arrays, pointers, structures and file handling for solving real world problems	Apply (Level 3)

Note: Attempt all the Questions.

Q1[CO1 (Understanding), 4 marks] Design a flowchart to find the sum of prime numbers in a close interval of [b, e] where b and e are integers.

Q2[CO2 (Understanding), 4 marks] A company wants to distribute an annual bonus among its employees. The total bonus is a fixed amount and the employees are divided into 3 groups: Managers, developers and interns. Bonuses must be distributed such that managers gets 50% of the total bonus, developers get 40% and interns get 10%. However, if the total bonus is not perfectly divisible, the left-over amount should be given to the group with the most members. Write a C program to calculate the bonuses for each group. Include validations for case like no employees or a zero-bonus pool.

Q3[CO2 (Understanding), 5 marks] Write a C program to print next greater elements in a given unsorted array. Elements for which no greater element exist, consider next greater element as -1.

Expected Output :

The given array is 5 3 10 9 6 13

Next bigger elements are :

Next bigger element of 5 in the array is : 10

Next bigger element of 3 in the array is : 10

Next bigger element of 10 in the array is : 13

Next bigger element of 9 in the array is : 13

Next bigger element of 6 in the array is : 13

Next bigger element of 13 in the array is : -1

Next bigger elements array is : 10 10 13 13 13 -1

Q4[CO3 (Understanding), 6 (1+2+1+1+1) marks] Alien communication points are tracked by (x, y). A coordinate is considered communicable if the product of its divisors (excluding itself) for x and y is the same for both values.

Tasks:

- Create a structure coordinate to store (x, y).
- Write a function to calculate the product of divisors.
- Check if any communicable points exist and display them.
- Allow dynamic input for multiple coordinates and check for rare communicable points where both x and y are prime.
- Use a union to store either the product of divisors or the prime status for each coordinate.

Q5[CO4 (Applying), 4 marks] Apply the concept of recursion to perform pattern printing for arbitrary value of number of rows. For number of rows = 3, pattern is:

```
1
2   3
4   5   6
```

Q6[CO5 (Applying), 6 marks] Develop a C program that dynamically allocates memory to store a string input by the user. The program should prompt the user to specify the maximum length of the string ensuring that the input data does not exceed the allocated memory. After obtaining the string, the program must count the occurrences of each alphabetic character (from 'a' to 'z') in the string, treating uppercase and lowercase letters as equivalent. The program should then display the frequency of each character. To make the program modular and organized, break the task into the following functions:

1. inputstring()
2. countCharfreq()
3. displayFreq()

Ensure proper memory management by freeing the allocated memory once it is no longer needed. Implement the solution in an iterative function-based approach.

Q7[CO5 (Applying), 6 marks] The IFSC code is an 11-character alphanumeric identifier made up of three parts: bank identifier, followed by a zero as a separator, and finally a unique 6-digit branch-code :

1. **Bank Code:** The first four alphabetic characters represent the bank like "ICIC" for ICICI bank.

2. **Zero:** The fifth character is always a '0', reserved as a separator for future use.

3. **Branch Code:** The final six characters uniquely identify the specific branch.

For example, IFSC for Langford Road branch of ICICI bank is ICIC0000269, where ICIC represents the bank name, 0 is for future use and 000269 denotes the branch code. A file named BankRecord contains 1000 such IFSC codes written using the following structure:

```
struct IFSC
{
    char BankCode;
    int separator;
    int BranchCode;
};
```

Write a C program to perform the following tasks:

- a) Separate the IFSC codes belonging to Axis bank and ICICI bank and write them back to two separate files named AxisBranch and ICICIBranch
- b) Count the number of SBIN branches stored in BankRecord file.